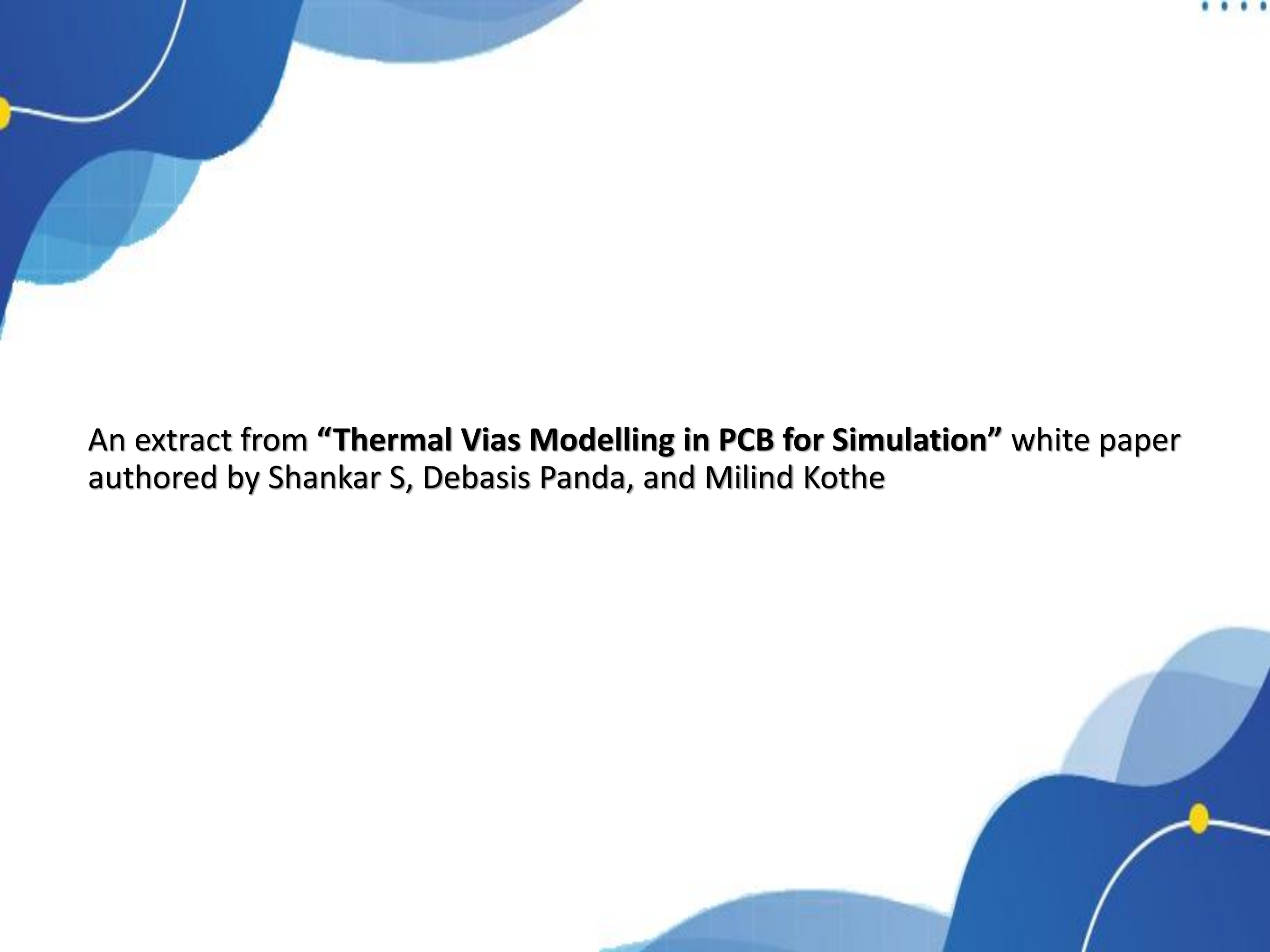




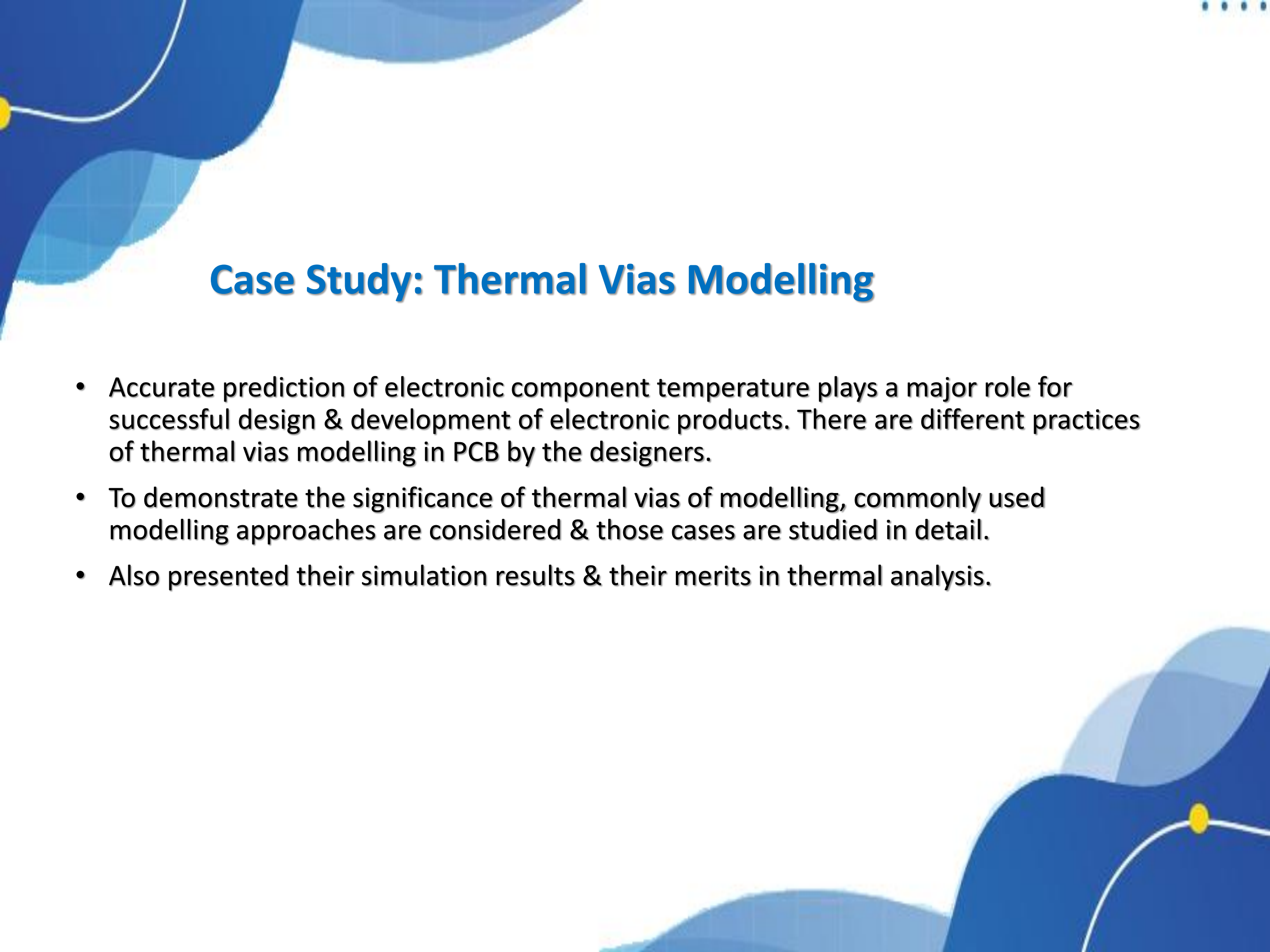
Shiretechnik Solutions Pvt Ltd

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Case Study on Thermal Vias Modeling for Simulation



An extract from “**Thermal Vias Modelling in PCB for Simulation**” white paper
authored by Shankar S, Debasis Panda, and Milind Kothe



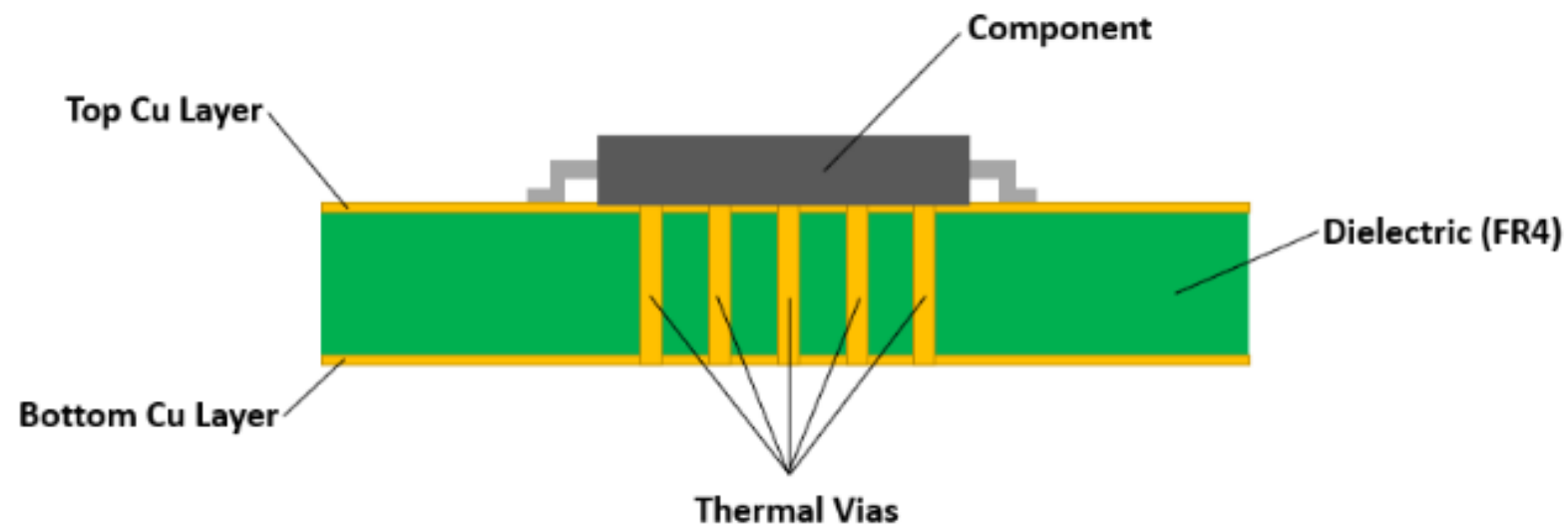
Case Study: Thermal Vias Modelling

- Accurate prediction of electronic component temperature plays a major role for successful design & development of electronic products. There are different practices of thermal vias modelling in PCB by the designers.
- To demonstrate the significance of thermal vias of modelling, commonly used modelling approaches are considered & those cases are studied in detail.
- Also presented their simulation results & their merits in thermal analysis.

Overview of Thermal Vias

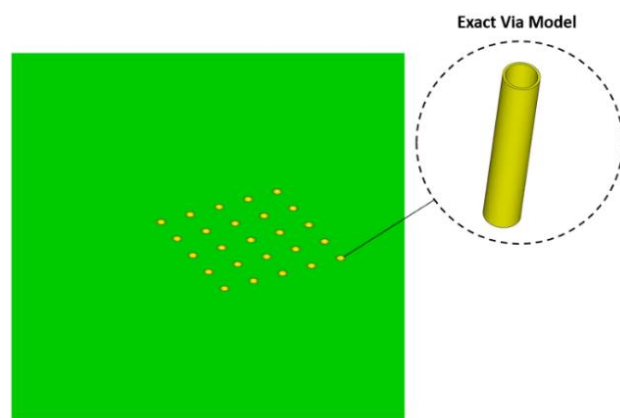
Thermal vias are plated through holes (PTHs) that are provided below the heat-source components in PCB.

These vias establish a significant heat path from the top copper layer to the bottom side of the PCB through lowering the through-plane thermal resistance of PCB under the component.

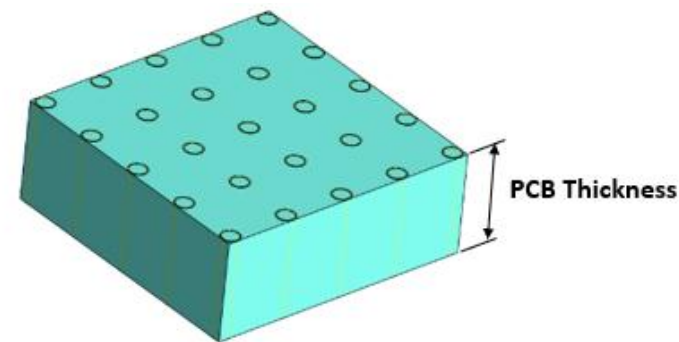


Four Different Modelling Approaches are considered

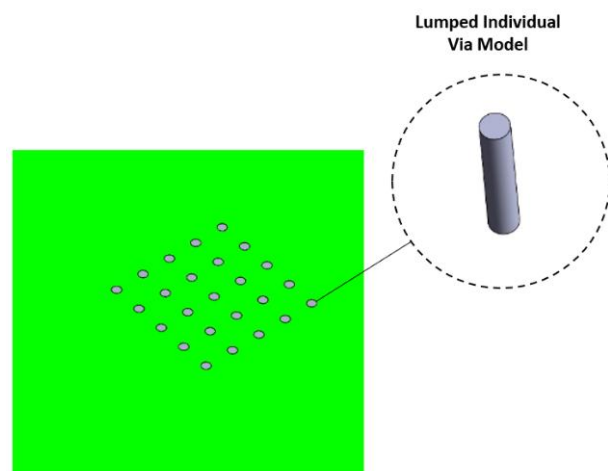
Exact Via Model



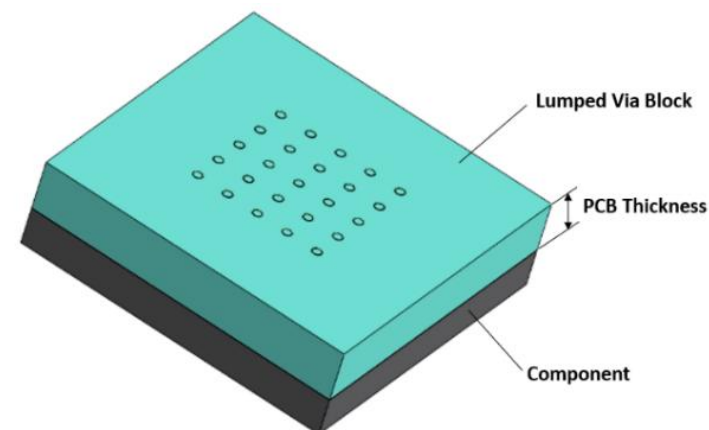
Lumped Vias Array Model



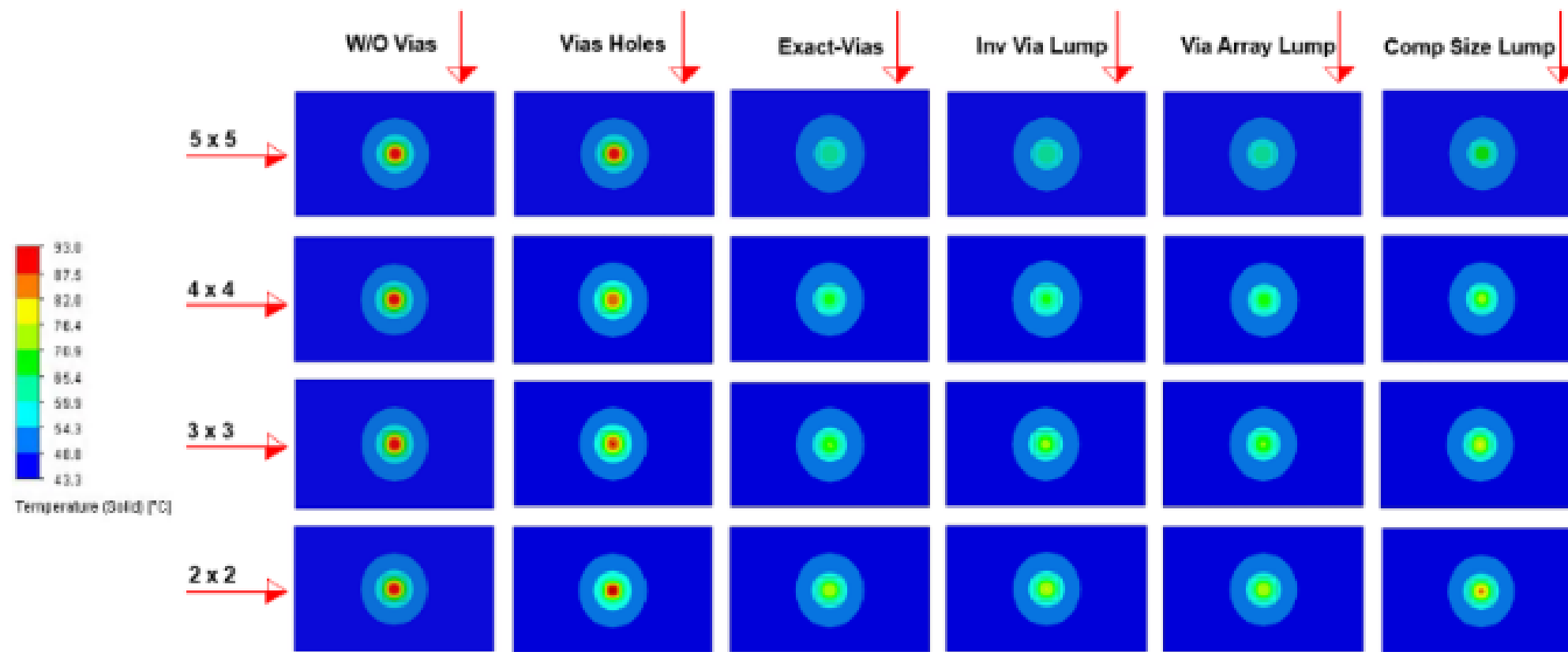
Lumped individual vias model



Lumped Component Size Model



Simulation Results: Natural Cooling Mode (1/2)

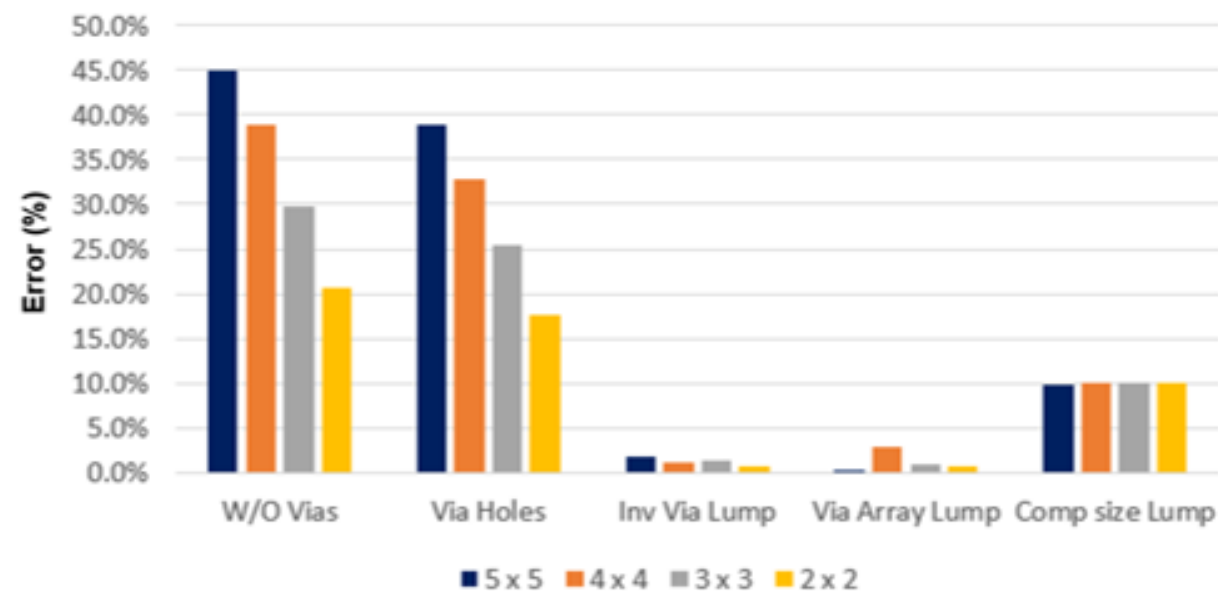


Temperature Contours of all modelling approaches

Simulation Results: Natural Cooling Mode (2/2)

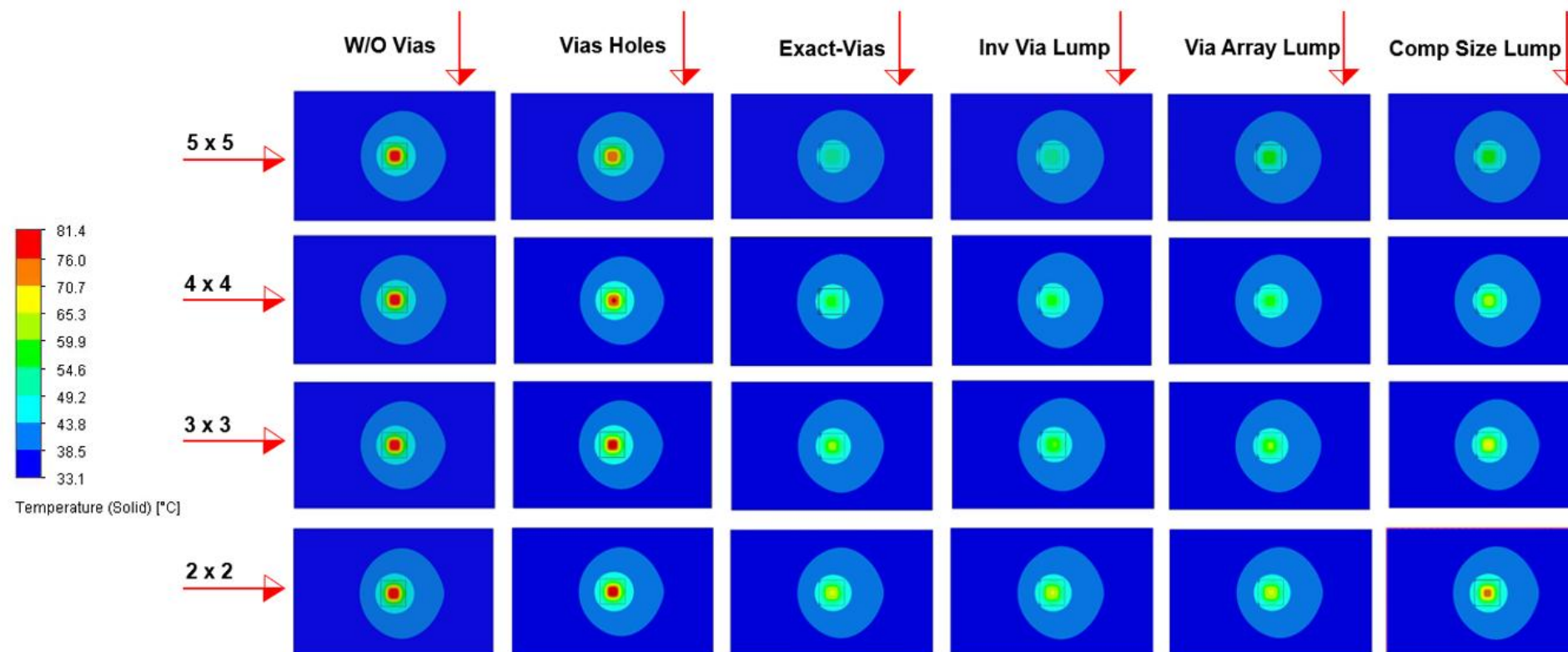
Array Size	Modelling Approach					
	W/O Vias	Via Holes	Exact Vias	Lumped Individual Vias	Lumped Via-Array	Lumped Component Size
5 x 5	93.0 °C	89.0 °C	64.1 °C	65.2 °C	64.3 °C	70.4 °C
4 x 4		89.0 °C	67.0 °C	67.7 °C	68.9 °C	73.7 °C
3 x 3		90.0 °C	71.7 °C	72.7 °C	72.3 °C	78.9 °C
2 x 2		90.7 °C	77.1 °C	77.7 °C	77.6 °C	84.8 °C

Simulated Component Temperatures of all modelling approaches



Simulated Temperature Error Against "Exact Vias Model"

Simulation Results: Forced Cooling Mode (1/2)



Temperature Contours of all modelling approaches

Simulation Results: Forced Cooling Mode (2/2)

Array Size	Modelling Approach					
	W/O Vias	Via Holes	Exact Vias	Lumped Individual Vias	Lumped Via-Array	Lumped Component Size
5 x 5	81.4 °C	75.8 °C	53.6 °C	53.2 °C	54.1 °C	60.0 °C
4 x 4		78.2 °C	56.6 °C	55.9 °C	56.9 °C	63.1 °C
3 x 3		80.1 °C	61.1 °C	60.7 °C	60.5 °C	68.1 °C
2 x 2		80.6 °C	66.7 °C	63.8 °C	66.5 °C	73.6 °C

Table 3: Simulated Component Temperatures of all modelling approaches

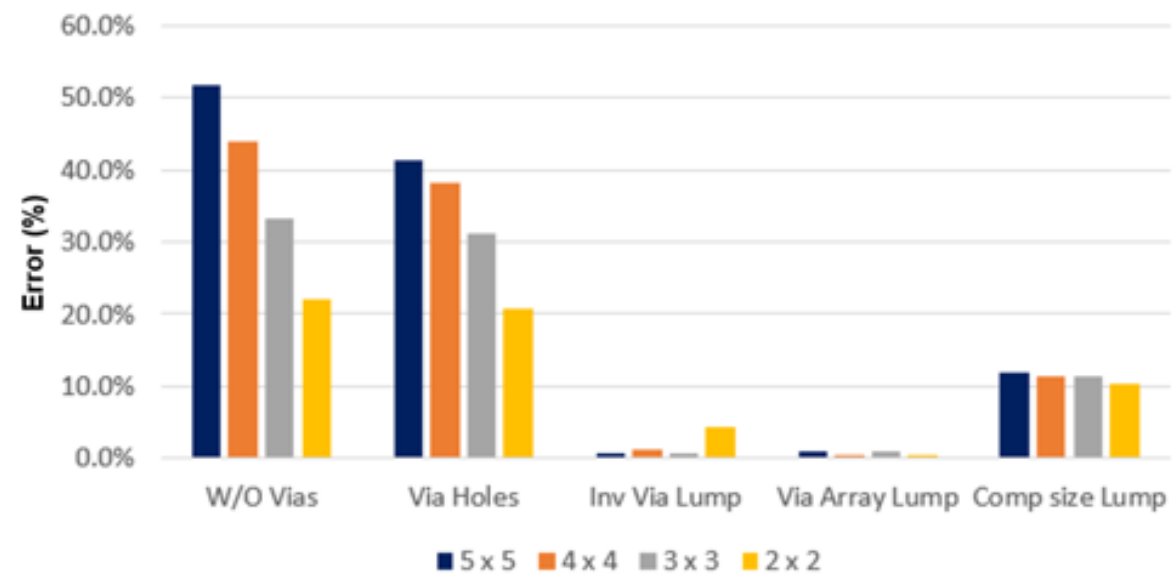


Chart 2: Simulated Temperature Error Against "Exact Vias Model"

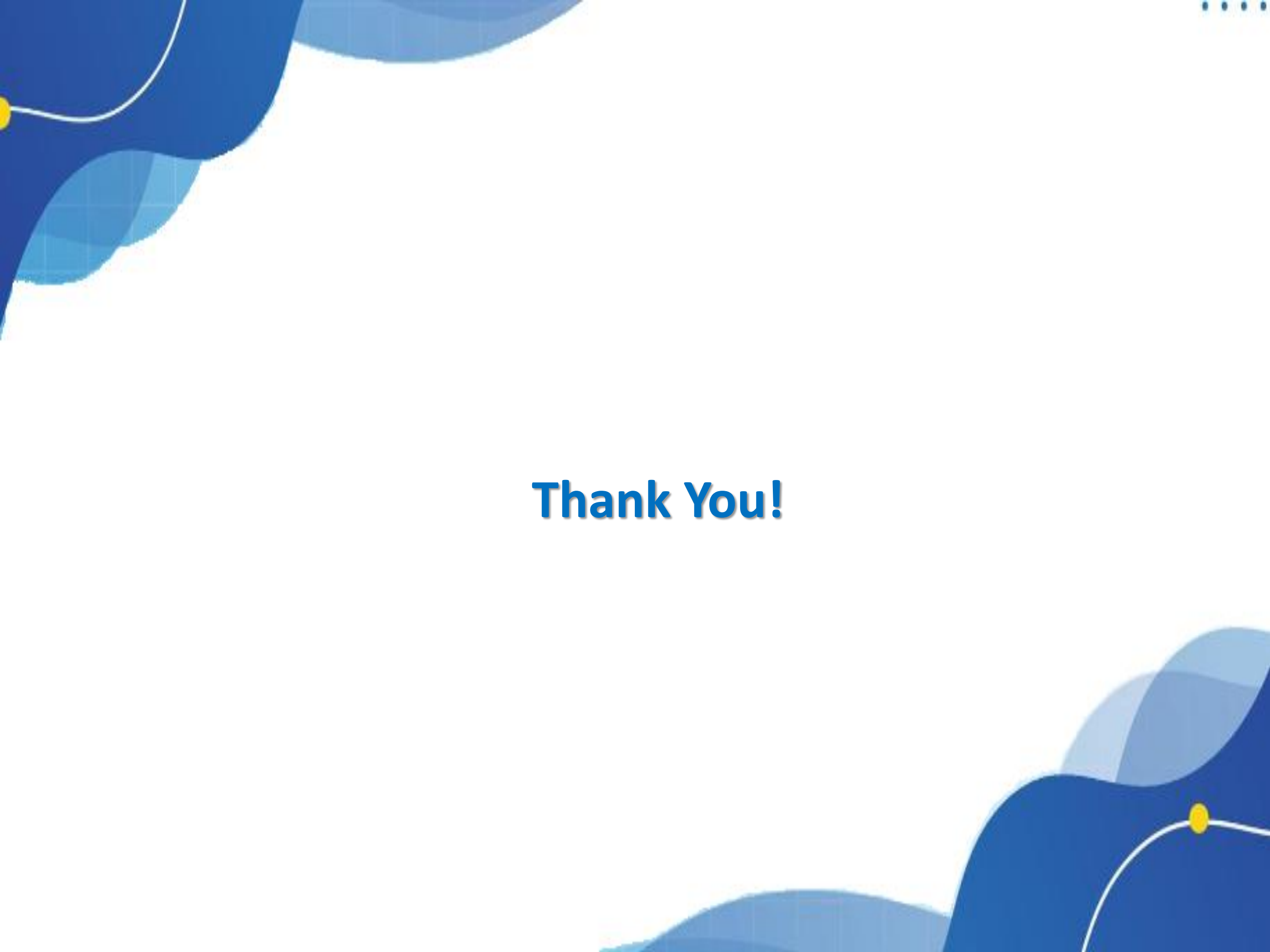
Conclusion

- It can be noted from the above simulation studies for both natural & forced cooling methods with different via array sizes that the ***“Lumped Individual Vias Model” & “Lumped Vias Array Model” consistently showed good agreements with “Exact Vias Model”***, not only the component temperature prediction & also the thermal images are closely matched.
- It can be concluded based on the simulation studies conducted that as “Lumped Vias Array Model” is quicker to model than other approaches discussed ***and it is also computationally efficient, this model can be well employed for Thermal Vias Modelling.***

Reference

- S. Shankar, P. Debasis and K. Milind ,White paper on “Thermal Vias Modelling in PCB for Simulation”, Shiretechnik Solutions Private limited Bangalore
- Nadezhda Kafadarova and Anna Andonova “PCB Thermal Design Improvement Through Thermal Vias”, Department of Microelectronics, Technical University of Sofia, Bulgaria.
- S. Lee, T.F. Lemczyk and M.M. Yovanovich, “Analysis of Thermal Vias in High Density Interconnect Technology”, Microelectronics Heat Transfer Laboratory, Department of Mechanical Engineering, University of Waterloo, Ontario, Canada.

*All the simulations are carried out using **Flow-simulation** software from **Solidworks** from **Dassault Systèmes***



Thank You!